



Engineered physics features (per leg; incorporate domain knowledge of a held-back leg)

thigh_tau_down $= \max(0, -\text{tau_thigh})$

retains only resistive (downward) thigh torque; a snag pulls the leg down against the motor

thigh_down_effort $= \text{thigh_tau_down} / (|\text{dq_thigh}| + 0.05)$

high torque with little motion = motor working hard but joint barely moving (the signature of a snag)

tau_sum $= \text{sum}_j |\text{tau}_j|, j \text{ in } \{\text{hip}, \text{thigh}, \text{calf}\}$

total per-leg effort regardless of direction; supports which-leg attribution

Why 0.40 s windows?

200 samples x 2 ms = 400 ms $\approx$ one trot gait stride (swing + stance). A single instant cannot separate a snag from normal gait variation; one stride carries the discriminative temporal context.

Window is positive if $\geq 50\%$ of its rows are Entangled; ambiguous (0.2-0.5) held out of training.